

Q4 .A. List three broad classifications of external, or peripheral, devices. 06

B. When a device interrupt occurs, how does the processor determine which device issued the interrupt? 06

C. What is the difference between programmed-driven I/O and interrupt-driven I/O? Give one advantage and one disadvantage of each. 06

Q5. A. Describe the classification of pipeline processors. 06

B. Give the principle of linear pipelining with the basic structure of a linear pipeline processor and the space-time diagram. 06

C. The diagram shown below is a reservation table for a unifunctional pipeline system. Identify a collision vector and draw a state diagram for this pipeline system. Also, find out the minimum average latency for the given pipeline system. 06

	t_0	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8
1	X								X
2		X	X					X	
3				X					
4					X	X			
5							X	X	